

Geometric Background

ROBOTICS — Lesson 04

Pose of a Rigid Body in 3D Euclidean Space

Flashback: 2D recap

- Recall the 2D case. Two frames with the same origin: frame O' — $x'y'$ is rotated by α . Point P has polar coordinates (ρ, θ) with respect to the $O - xy$ frame.

- Applying simple rules on the trigonometric functions:

$$p_{x'} = p_x \cos(\alpha) + p_y \sin(\alpha)$$

$$p_{y'} = -p_x \sin(\alpha) + p_y \cos(\alpha)$$

which means $p' = R(\alpha)p$, with

$$R(\alpha) = \begin{pmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{pmatrix}$$

- $R(\alpha)$ is called a rotation matrix.

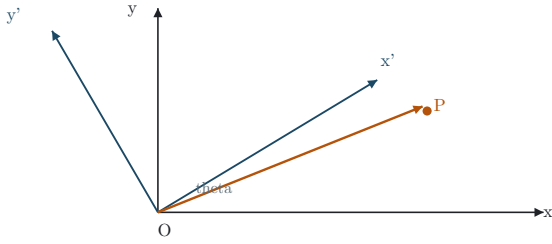


Fig. 1: Frame $O' - x'y'$ rotated by α ; point P with polar coordinates (ρ, θ) in $O - xy$.

Pose of a Rigid Body

- Robots are modelled as a collection of rigid bodies.
- Preliminary point: define what we mean by **pose**.
- Generic rigid body: assume a frame attached to the body. By the definition of a rigid body, the coordinates of its points are invariant in the frame attached to the robot.

Frame identification

The frame is uniquely identified by:

- the position of its origin O' (which is a bound vector) $\vec{o}' = \begin{pmatrix} o_{x'} \\ o_{y'} \\ o_{z'} \end{pmatrix}$;
- the coordinates of the three unit vectors x', y', z' of the frame attached to the robot, expressed in the world frame:

$$x' = x'_x x + x'_y y + x'_z z$$

$$y' = y'_x x + y'_y y + y'_z z$$

$$z' = z'_x x + z'_y y + z'_z z$$

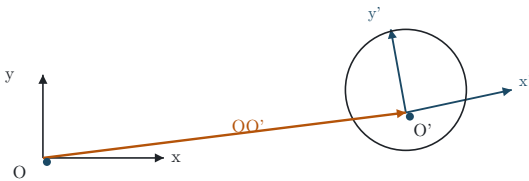


Fig. 2: Two frames and the vector joining the origins O' .

Coordinates through the scalar product

Let P be the vector joining O with a generic point of the body, P' the vector joining O' with the same point, and $\mathbb{O}'$ the vector joining the two origins. Then:

$$P = P' + \mathbb{O}'$$

Given P and the unit vectors x, y, z , the coordinates of P are:

$$p_x = P \cdot x, \quad p_y = P \cdot y, \quad p_z = P \cdot z$$

where $\cdot$ stands for the scalar product.

Since $P = P' + \mathbb{O}'$ and $P' = p_{x'}x' + p_{y'}y' + p_{z'}z'$:

$$p_x = P \cdot x = P' \cdot x + \mathbb{O}' \cdot x = p_{x'}(x' \cdot x) + p_{y'}(y' \cdot x) + p_{z'}(z' \cdot x) + \mathbb{O}' \cdot x$$

$$p_y = P \cdot y = P' \cdot y + \mathbb{O}' \cdot y = p_{x'}(x' \cdot y) + p_{y'}(y' \cdot y) + p_{z'}(z' \cdot y) + \mathbb{O}' \cdot y$$

$$p_z = P \cdot z = P' \cdot z + \mathbb{O}' \cdot z = p_{x'}(x' \cdot z) + p_{y'}(y' \cdot z) + p_{z'}(z' \cdot z) + \mathbb{O}' \cdot z$$

In simple terms:

$$\begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = R \begin{pmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{pmatrix} + \begin{pmatrix} \mathbb{O}'_x \\ \mathbb{O}'_y \\ \mathbb{O}'_z \end{pmatrix}$$

with

$$R = \begin{pmatrix} x'_x & y'_x & z'_x \\ x'_y & y'_y & z'_y \\ x'_z & y'_z & z'_z \end{pmatrix}$$

or equivalently

$$R = \begin{pmatrix} x'^T x & y'^T x & z'^T x \\ x'^T y & y'^T y & z'^T y \\ x'^T z & y'^T z & z'^T z \end{pmatrix}$$

Rotation Matrices

Rotation Matrix

- A matrix defined using the coordinates of the three unit vectors is called a **rotation matrix**.

Rotation Matrix

$$R = \begin{pmatrix} x'_x & y'_x & z'_x \\ x'_y & y'_y & z'_y \\ x'_z & y'_z & z'_z \end{pmatrix}$$

The different elements of R can be found by using the scalar product:

$$R = \begin{pmatrix} x' \cdot x & y' \cdot x & z' \cdot x \\ x' \cdot y & y' \cdot y & z' \cdot y \\ x' \cdot z & y' \cdot z & z' \cdot z \end{pmatrix}$$

Orthogonality

Let us compute RR^T :

$$RR^T = \begin{pmatrix} x' \cdot x' & y' \cdot x' & z' \cdot x' \\ x' \cdot y' & y' \cdot y' & z' \cdot y' \\ x' \cdot z' & y' \cdot z' & z' \cdot z' \end{pmatrix}$$

Since the frame $O' - x'y'z'$ is made of orthonormal vectors, $x' \cdot x' = y' \cdot y' = z' \cdot z' = 1$, while $x' \cdot y' = x' \cdot z' = y' \cdot z' = 0$, which leads to:

$$R^T R = I_3 \rightarrow R^T = R^{-1}$$

Key Property of the Rotation Matrix The Rotation matrix is an orthogonal matrix with $\det(R) = 1$ for right-handed frames and $\det(R) = -1$ for left-handed frames.

SO(3)

Rotation matrices have the following properties:

- Composing (i.e., multiplying) two rotation matrices produces another rotation matrix.
- Rotations respect the associative property: $R_1(R_2 R_3) = (R_1 R_2) R_3$.
- Every rotation has a unique inverse.
- The degenerate rotation I_3 is a rotation.
- Rotations do not commute.
- These properties qualify rotations as an algebraic non-abelian (i.e., non-commutative) group under composition, named $SO(3)$.

Elementary Rotations

Rotation by an angle α about the z -axis:

$$x' = \begin{pmatrix} \cos(\alpha) \\ \sin(\alpha) \\ 0 \end{pmatrix}, \quad y' = \begin{pmatrix} -\sin(\alpha) \\ \cos(\alpha) \\ 0 \end{pmatrix}, \quad z' = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

with resulting matrix

$$R_z(\alpha) = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Likewise, for an angle β about the y -axis and an angle γ about the x -axis:

$$R_y(\beta) = \begin{pmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{pmatrix} \quad R_x(\gamma) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{pmatrix}$$

We can easily verify that:

$$R_k(-\theta) = R_k^T(\theta), \quad k = x, y, z$$

Geometric Interpretation

Geometric Interpretation of a Rotation Matrix A rotation matrix R represents a rotation about an axis in space required to align the axes of the reference frame with the corresponding axes of the body frame.

Representation of a Vector

- Recap of the representation of a vector in the case where the two origins O and O' coincide.

- A generic point is represented by $p = \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix}$ in the $O - xyz$ frame and by $p' = \begin{pmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{pmatrix}$ in the $O - x'y'z'$ frame.

- Since both p and p' refer to the same point:

$$p = p_{x'}x' + p_{y'}y' + p_{z'}z' = (x'y'z')p' = Rp'$$

and exploiting the properties of R :

$$p' = R^T p$$

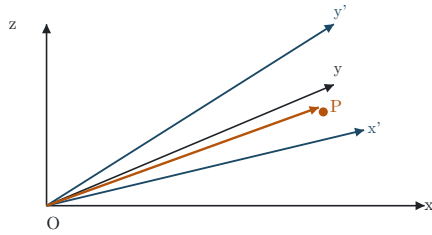


Fig. 3: Coincident origins: same point expressed in the two frames.

Rotation of a Vector

- Another way of seeing a rotation matrix: a matrix operator that enables the rotation of a vector about an arbitrary axis in space.
- Consider a rotation matrix R , a vector p' expressed in the $O - xyz$ frame and the vector $p = Rp'$. Then:

$$\|p\|^2 = p^T p = (p')^T R^T R p' = (p')^T p'$$

- The rotation operator preserves the norm.

Worked Examples

Coordinate transformation ($p' = R_{z(\alpha)}^T p$)

Generalisation to 3D of the 2D example. The frame $O' - x'y'z'$ is rotated by α about the z -axis; point P has polar coordinates (ρ, θ) with respect to $O - xyz$. It is easy to see that:

$$p_x = \rho \cos(\theta), \quad p_y = \rho \sin(\theta)$$

$$p_{x'} = \rho \cos(\theta - \alpha), \quad p_{y'} = \rho \sin(\theta - \alpha)$$

Applying simple trigonometric functions:

$$p_{x'} = p_x \cos(\alpha) + p_y \sin(\alpha)$$

$$p_{y'} = -p_x \sin(\alpha) + p_y \cos(\alpha)$$

$$p_{z'} = p_z$$

which means $p' = R_{z(\alpha)}^T p$.

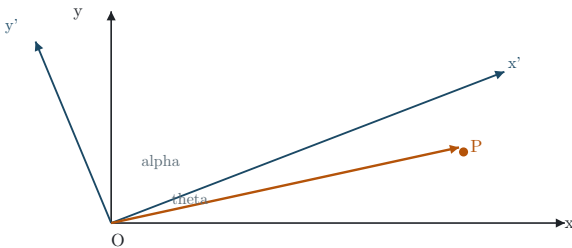


Fig. 4: Point P with polar coordinates (ρ, θ) in a frame rotated by α about z .

Rotation of a vector ($p = R_{z(\alpha)} p'$)

Consider the vector obtained by rotating a vector p' by an angle α about the z -axis; p' has polar coordinates (ρ, θ) . Vector p will have polar coordinates $(\rho, \alpha + \theta)$. We have:

$$p_{x'} = \rho \cos(\theta), \quad p_{y'} = \rho \sin(\theta)$$

$$p_x = \rho \cos(\theta + \alpha), \quad p_y = \rho \sin(\theta + \alpha)$$

$$p_z = p_{z'}$$

With a little effort, we find:

$$p_x = p_{x'} \cos(\alpha) - p_{y'} \sin(\alpha)$$

$$p_y = p_{x'} \sin(\alpha) + p_{y'} \cos(\alpha)$$

$$p_z = p_{z'}$$

which means $p = R_{z(\alpha)} p'$.

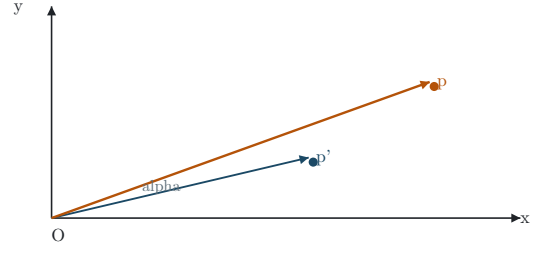


Fig. 5: Vector p' rotated by α about z gives p .

Summary

Geometric Meanings of a Rotation Matrix

- It describes the **mutual orientation** between two coordinate frames. Its column vectors are the direction cosines of the axes of the rotated frame with respect to the original one.
- It represents the **coordinate transformation** between two rotated frames.
- It is the **operator** that enables rotations within the same coordinate frame.